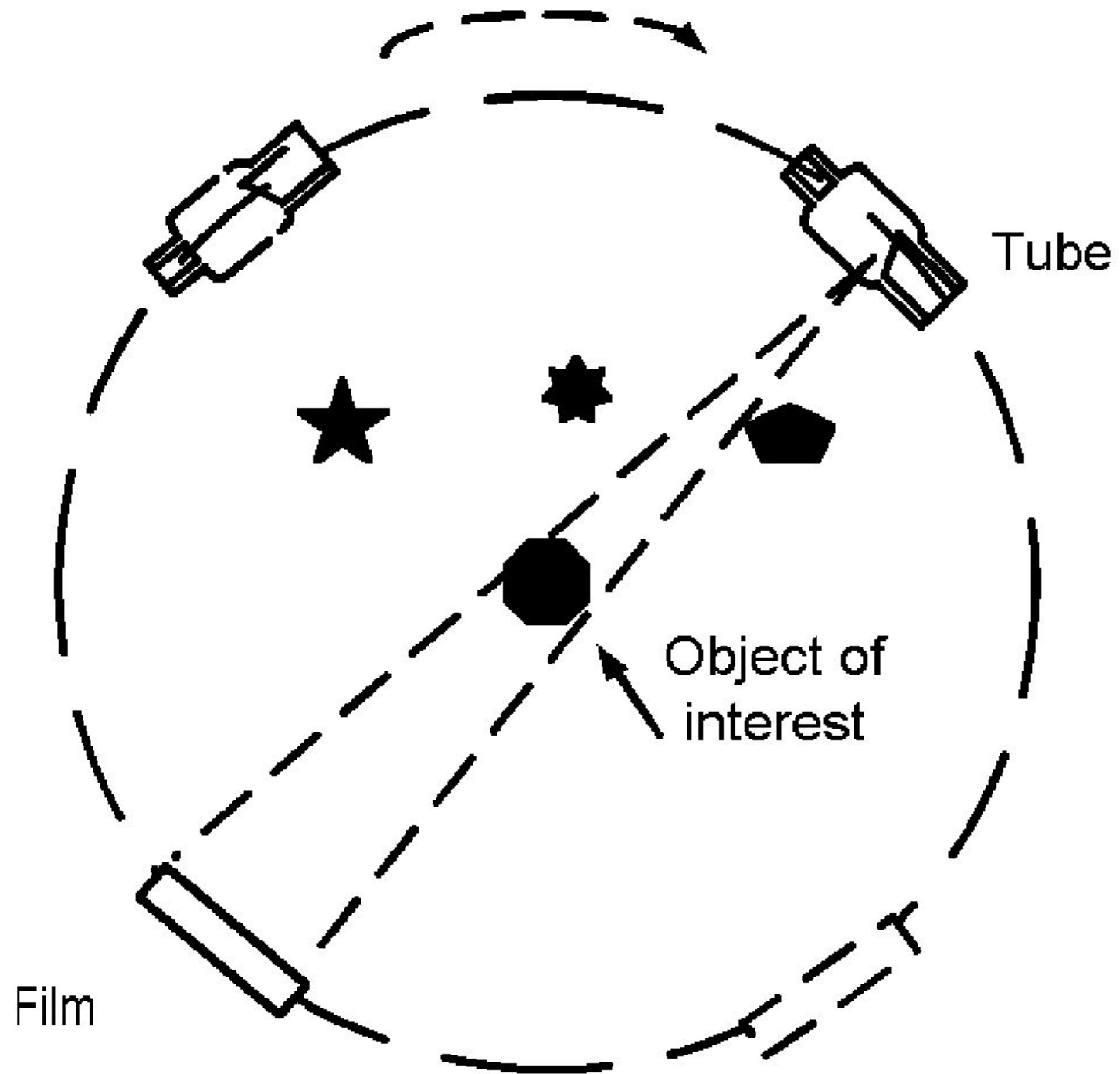


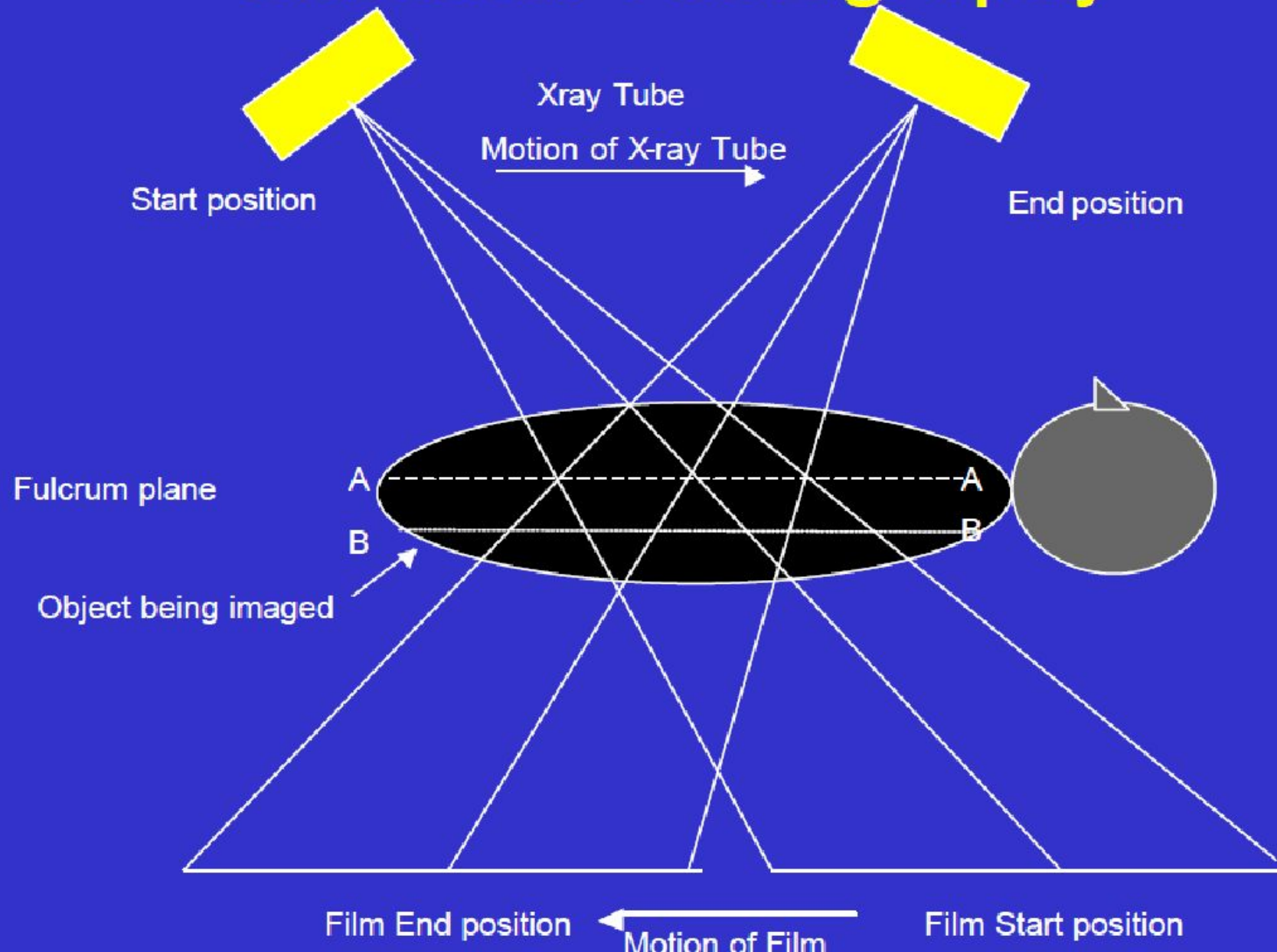
Reconstruction Algorithms-II

Renu John
BME

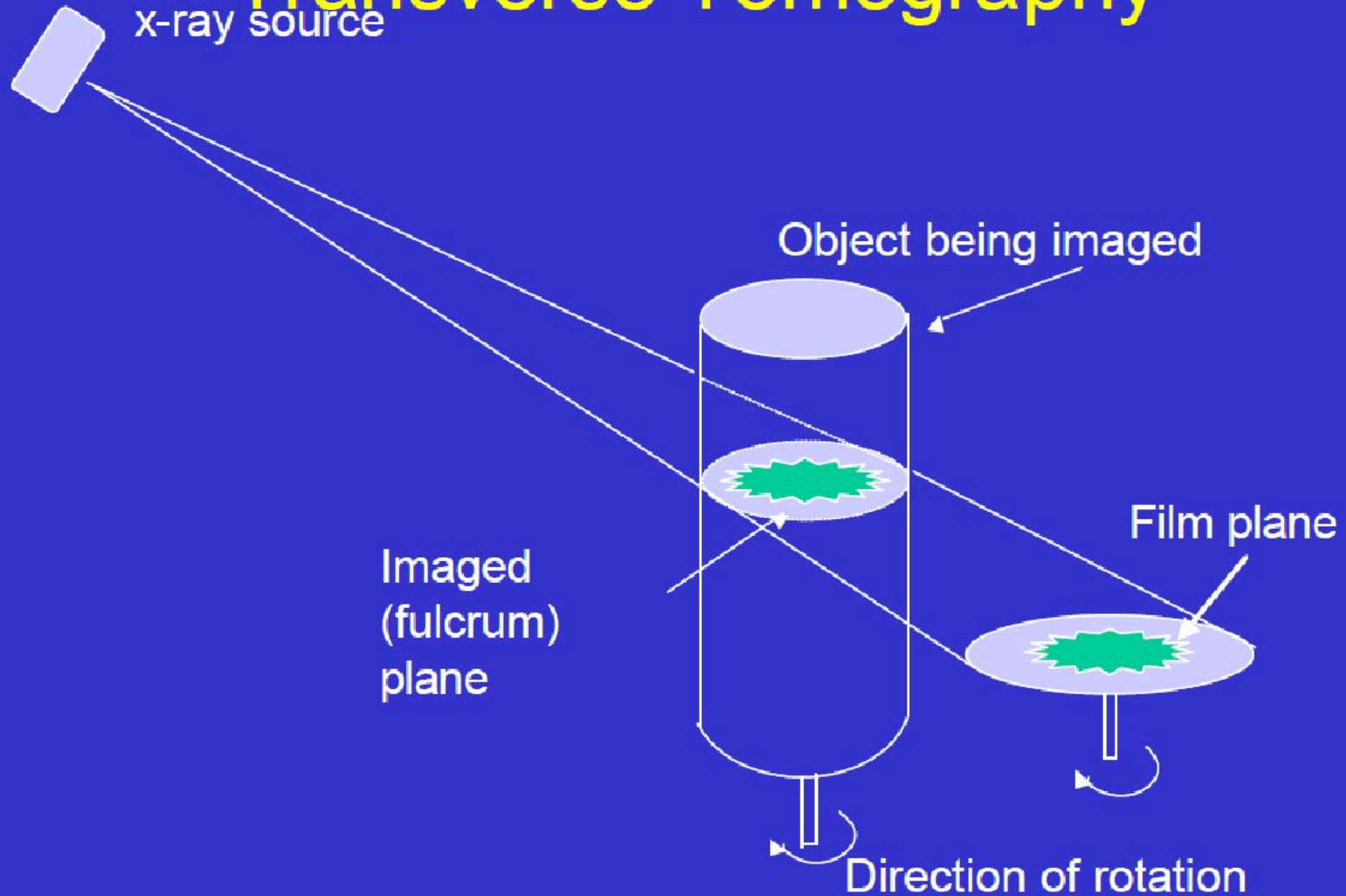
Mar 18th 2025

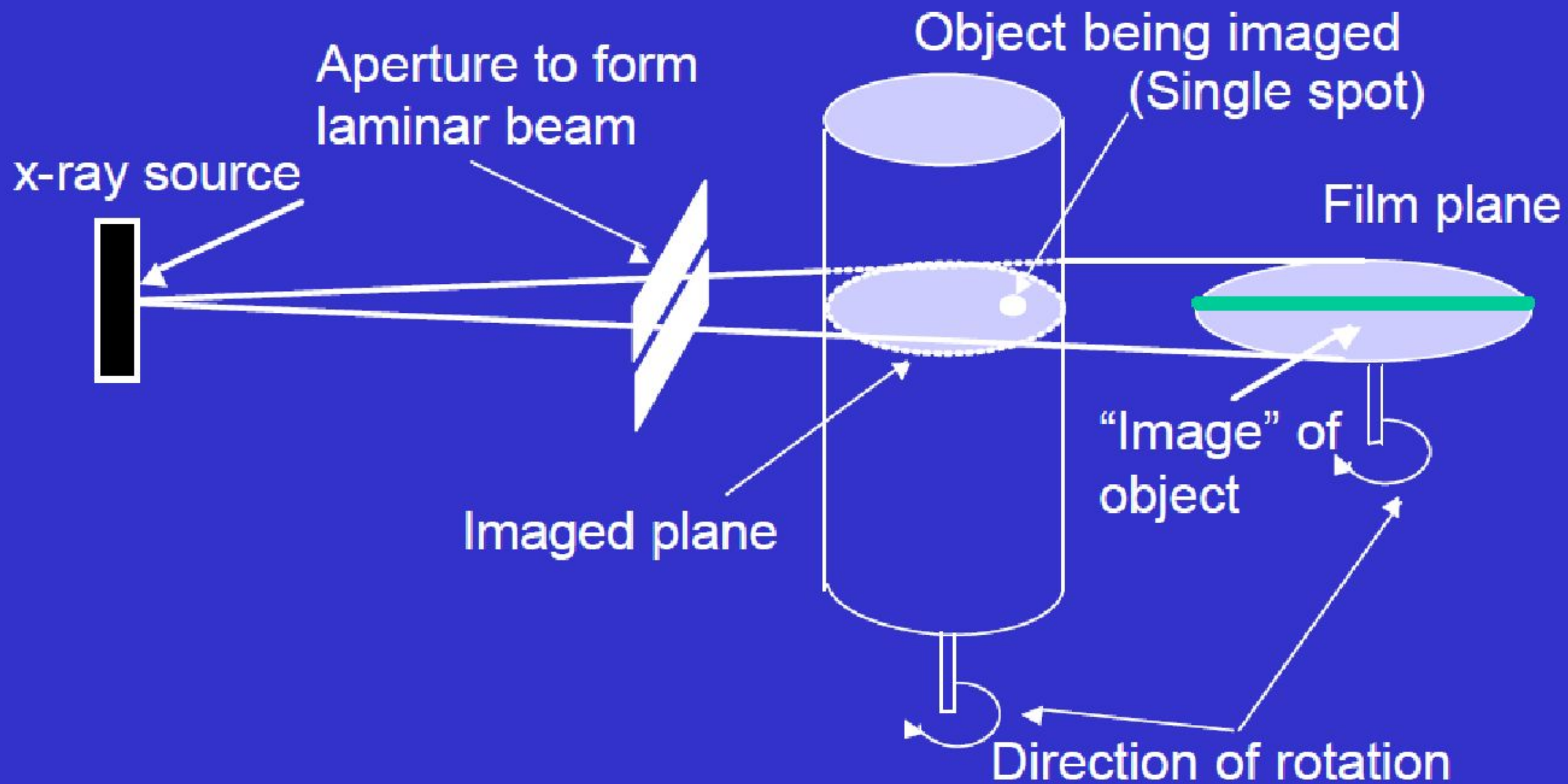


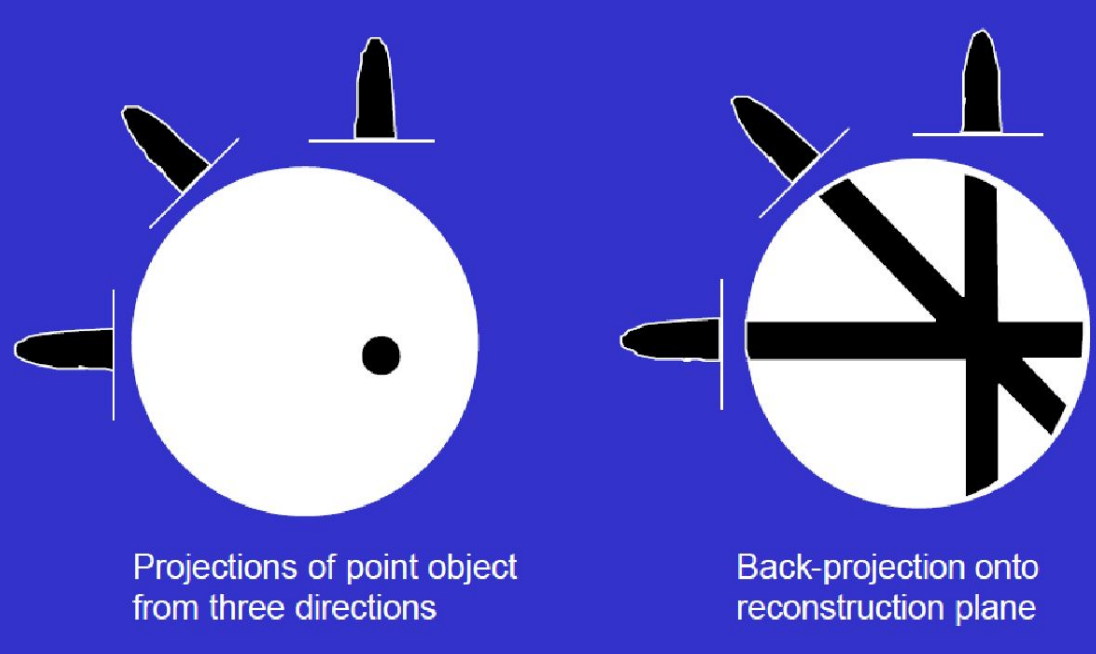
Classical Tomography



Transverse Tomography

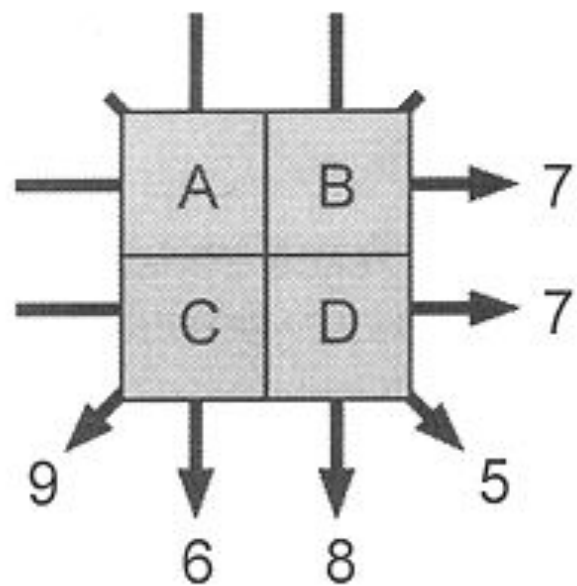






Projections of point object
from three directions

Back-projection onto
reconstruction plane



problem

$$\begin{aligned} A + B &= 7 \\ A + C &= 6 \\ A + D &= 5 \\ B + C &= 9 \\ B + D &= 8 \\ C + D &= 7 \end{aligned}$$

method

2	5
4	3

solution

$$f(x) = \int_{-\infty}^{\infty} F(u) e^{-i2\pi ux} du$$

“forward” Fourier Transform

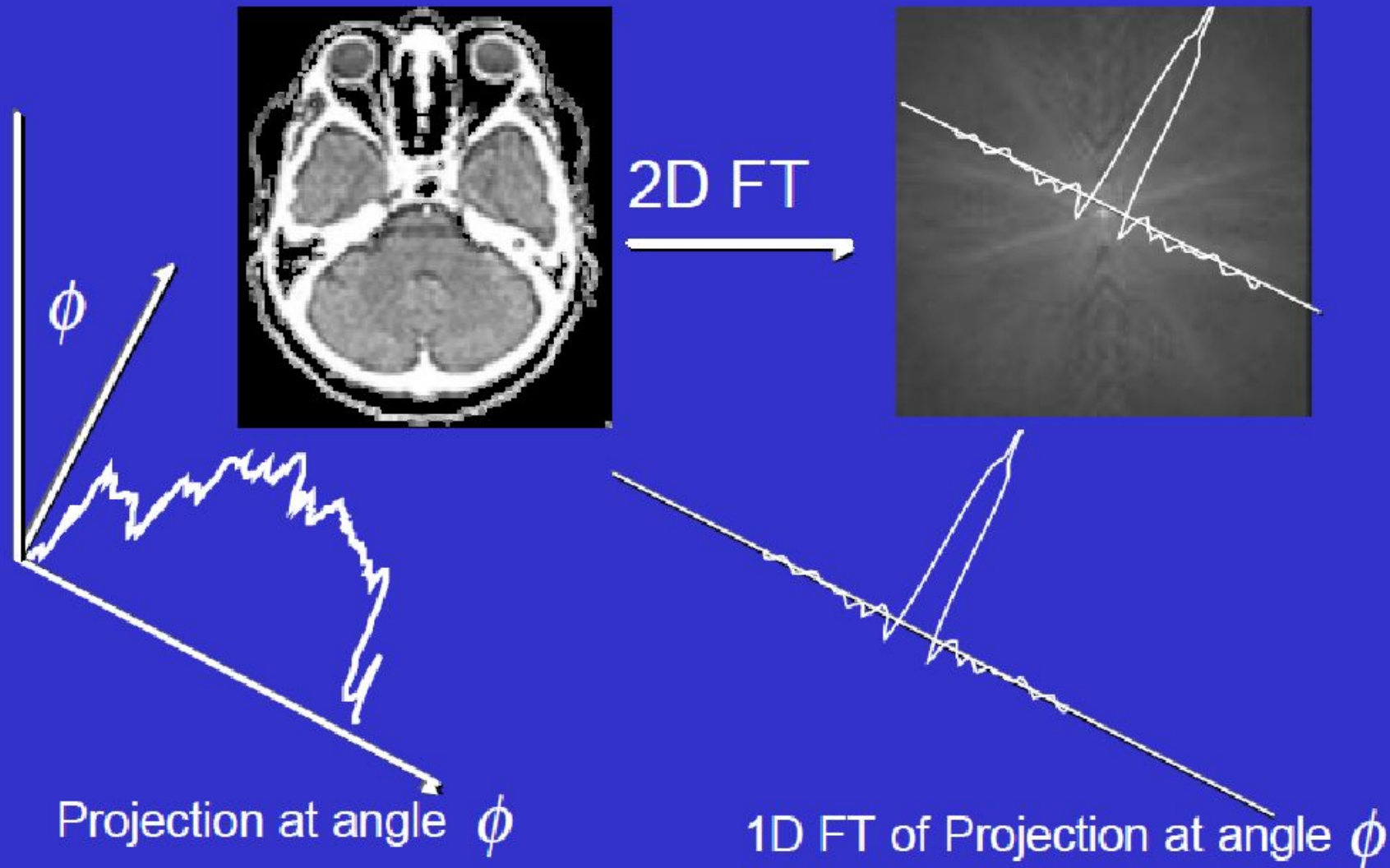
$$F(u) = \int_{-\infty}^{\infty} f(x) e^{i2\pi ux} dx$$

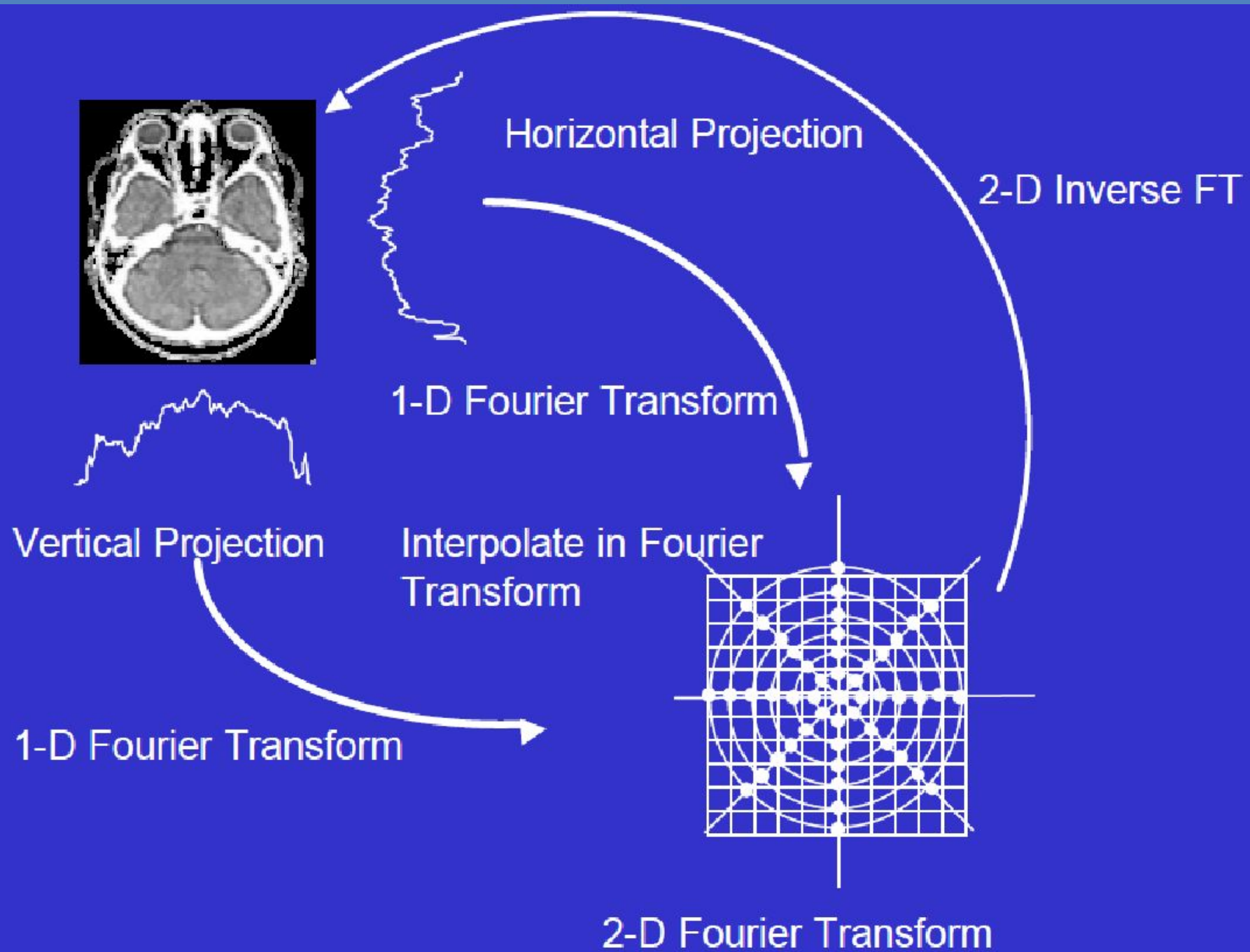
“inverse” Fourier Transform

Central Slice Theorem

- Pivotal to understanding of CT reconstruction
- Relates 2D FT of image to 1D FT of its projection
- N.B. 2D FT is “k-space” of MRI

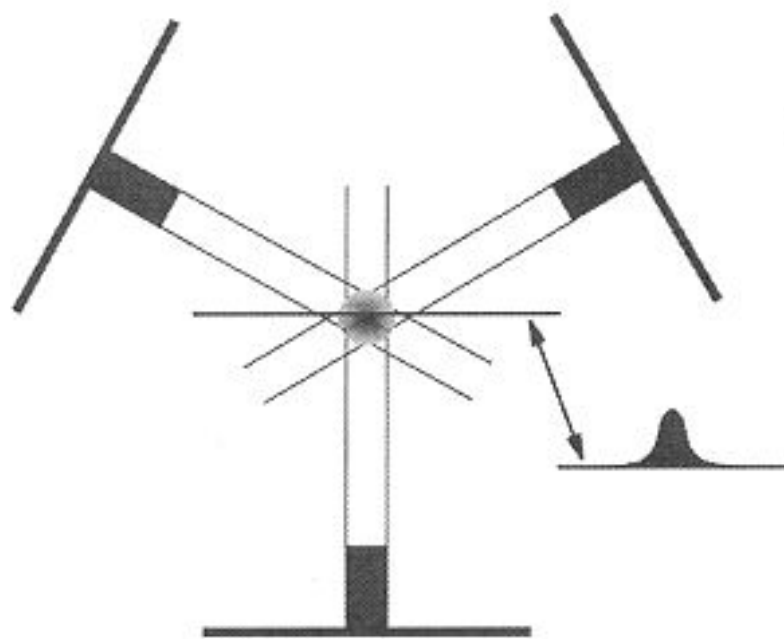
Central Slice Theorem



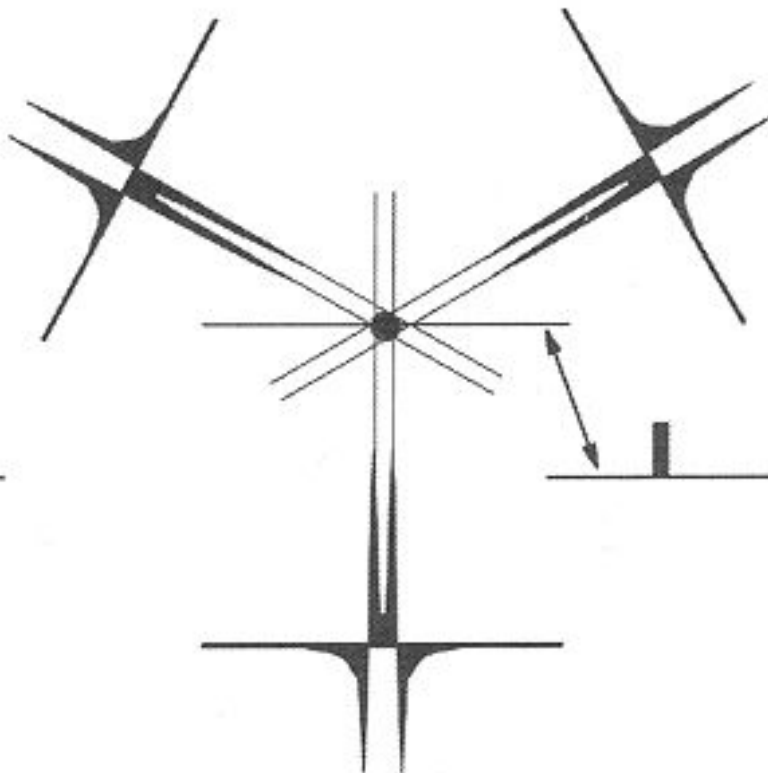


Simple backprojection

- Starts with an empty image matrix, and the μ value from each ray in all views is added to each pixel in a line through the image corresponding to the ray's path
- A characteristic $1/r$ blurring is a byproduct
- A filtering step is therefore added to correct this blurring



simple backprojection



filtered backprojection

$$\text{1D Rectangular Pulse} \otimes \text{1D Filter} = \text{2D Filtered Backprojection Result}$$

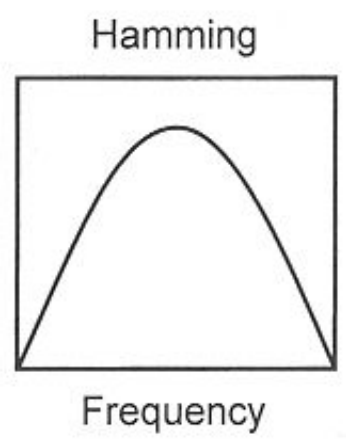
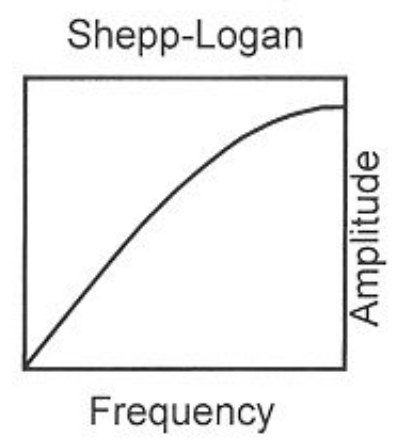
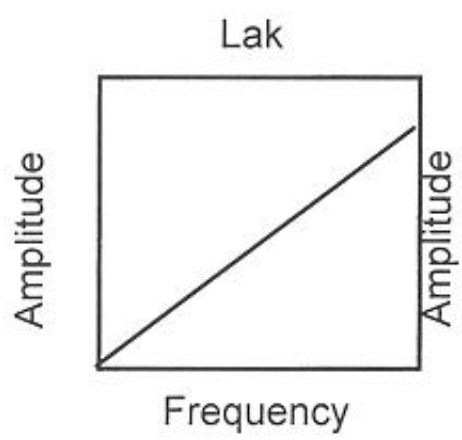
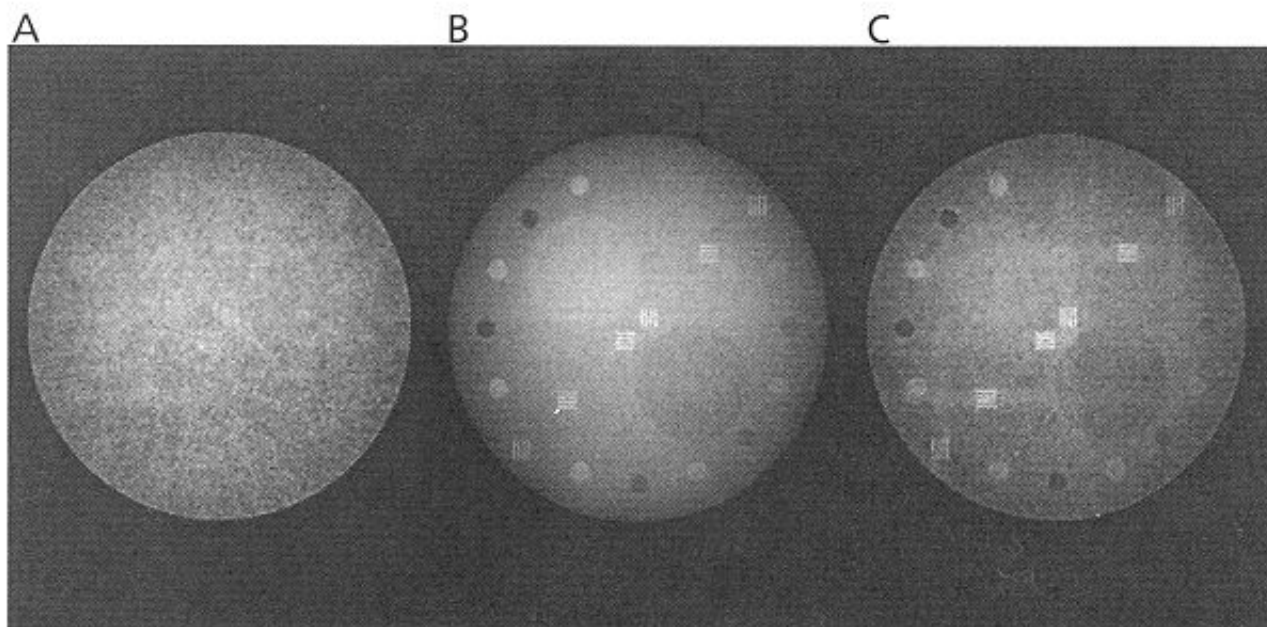
The equation shows the relationship between the 1D rectangular pulse (the original object), the 1D filter (the filter kernel), and the resulting 2D filtered backprojection image. The symbol $\otimes$ represents the 2D convolution operation.

Filtered backprojection

- The raw view data are mathematically filtered before being backprojected onto the image matrix
- Involves convolving the projection data with a convolution kernel
- Different kernels are used for varying clinical applications such as soft tissue imaging or bone imaging

Convolution filters

- Lak filter increases amplitude linearly as a function of frequency; works well when there is no noise in the data
- Shepp-Logan filter incorporates some *roll-off* at higher frequencies, reducing high-frequency noise in the final CT image
- Hamming filter has even more pronounced high-frequency roll-off, with better high-frequency noise suppression



Iterative reconstruction methods

Conventional iterative algebraic methods

Algebraic Reconstruction Technique (ART)

Simultaneous Iterative Reconstruction Technique (SIRT)

Iterative Least-squares Technique (ILST)

Iterative statistical reconstruction methods

(with and without using a priori information)

Gradient And Conjugate Gradient (CG) Algorithms

Maximum Likelihood Expectation Maximization (MLEM)

Ordered-subsets Expectation Maximization (OSEM)

Maximum A Posteriori (MAP) Algorithms

The principle of the iterative algorithms is to find a solution (that is - to reconstruct an image of a tomographic slice from projections) by *successive estimates*.

The projections corresponding to the current estimate are compared with the measured projections. The result of the comparison is used to modify the current estimate, thereby creating a new estimate.

The algorithms differ in the way the measured and estimated projections are compared and the kind of correction applied to the current estimate.

The process is initiated by arbitrarily creating a first estimate - for example, a uniform image (all pixels equal zero, one, or a mean pixel value,...).

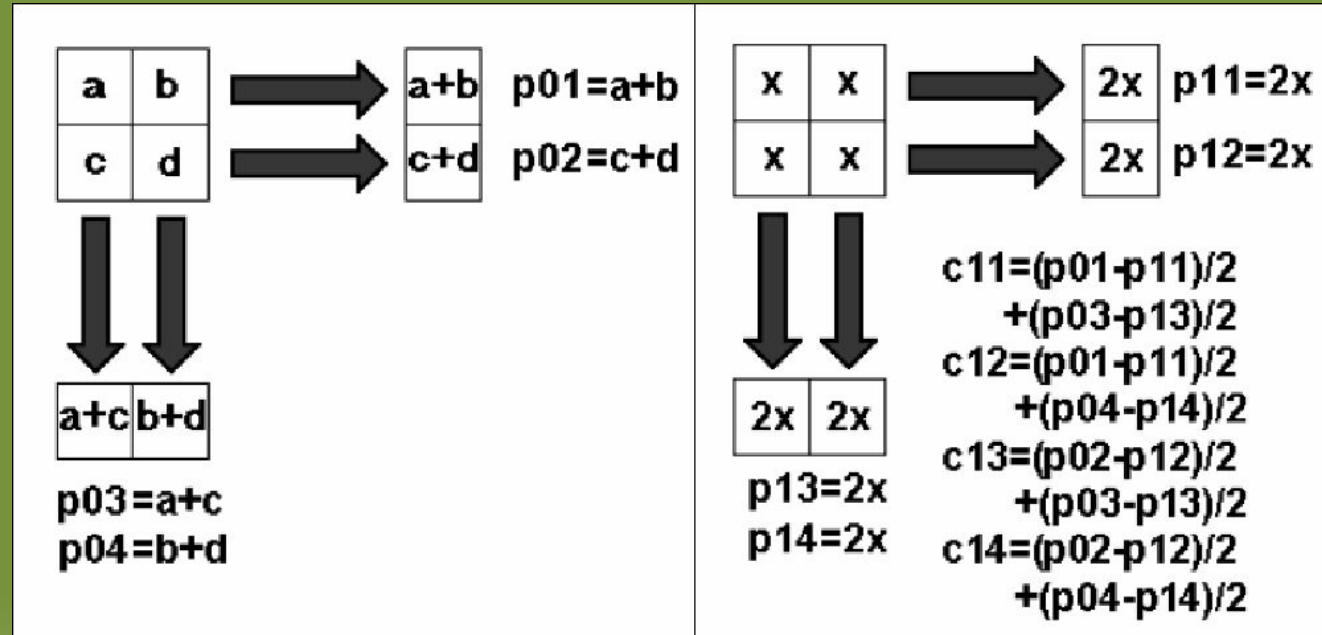
Corrections are carried out either as addition of differences or multiplication by quotients between measured and estimated projections.

Algorithm

- 1) Make *the first arbitrary estimate* of the slice (homogeneous image)
- (2) *Project the estimated slice* into projections analogous to those measured by the camera (***important:*** in this step, physical corrections can be introduced - for attenuation, scatter, and depth-dependent collimator resolution)
- (3) *compare the projections* of the estimate with measured projections (subtract or divide the corresponding projections in order to obtain correction factors - in the form of differences or quotients)
- (4) *stop or continue*: if the correction factors are approaching zero, if they do not change in subsequent iterations, or if the maximum number of iterations was achieved, then finish; otherwise
- (5) *apply corrections* to the estimate (add the differences to individual pixels or multiply pixel values by correction quotients) - thus make the new estimate of the slice
- (6) *go to step (2)*.

ART first algorithm

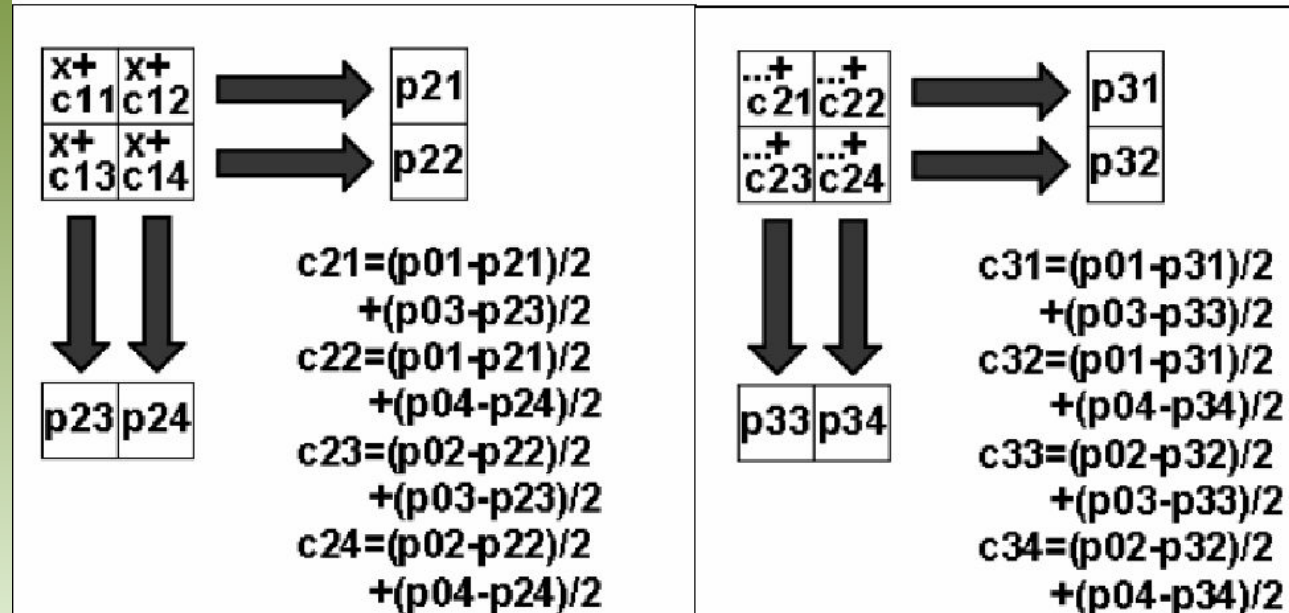
measured
projections



first estimate
and its
projections

$$x=(a+b+c+d)/4$$

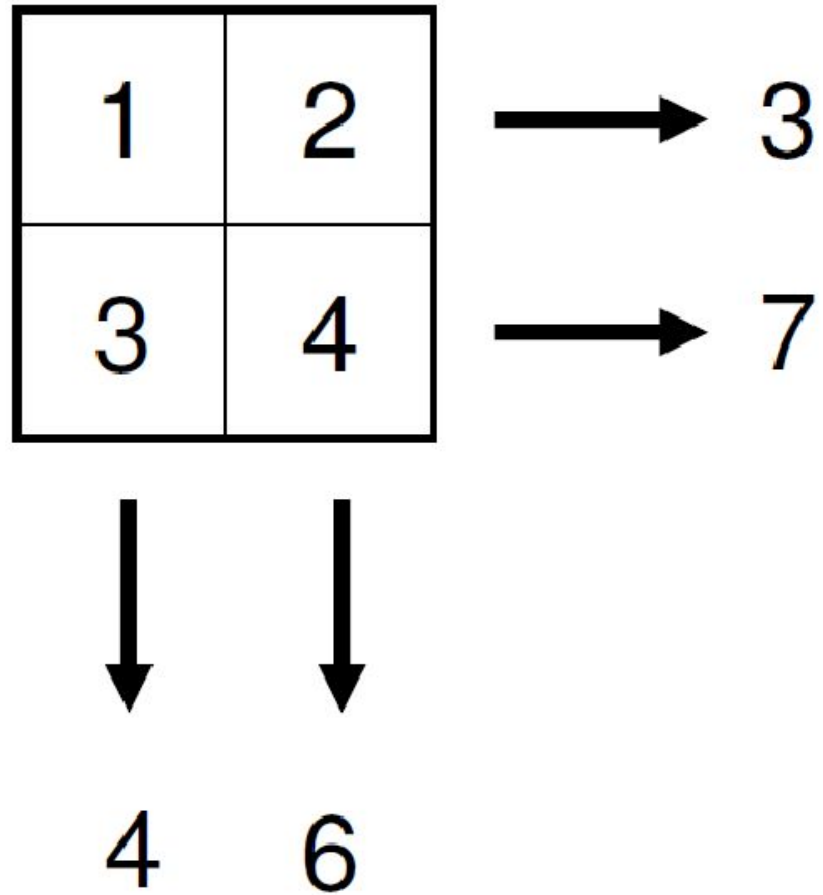
first
iteration
(additive
corrections)



second
iteration
(additive
corrections)

Example: Use the proposed ART algorithm

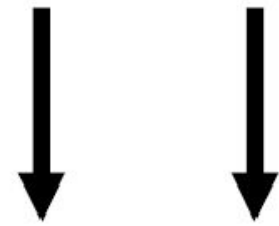
Original data



?	?
?	?

→ **3**

→ **7**

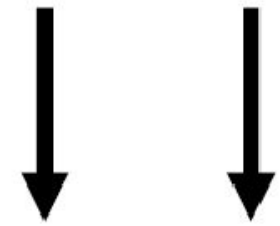


4 6

2.5	2.5
2.5	2.5

→ 5

→ 5



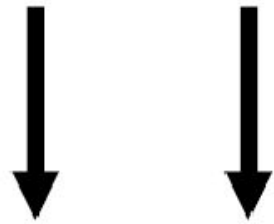
5 5

$$x = (\mathbf{3} + \mathbf{7}) / 4 = 10 / 4 = \mathbf{2.5}$$

?	?
?	?

→ **3**

→ 7

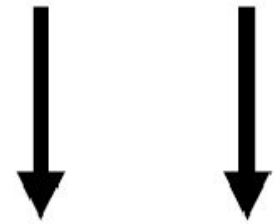


4 6

2.5	2.5
2.5	2.5

→ **5**

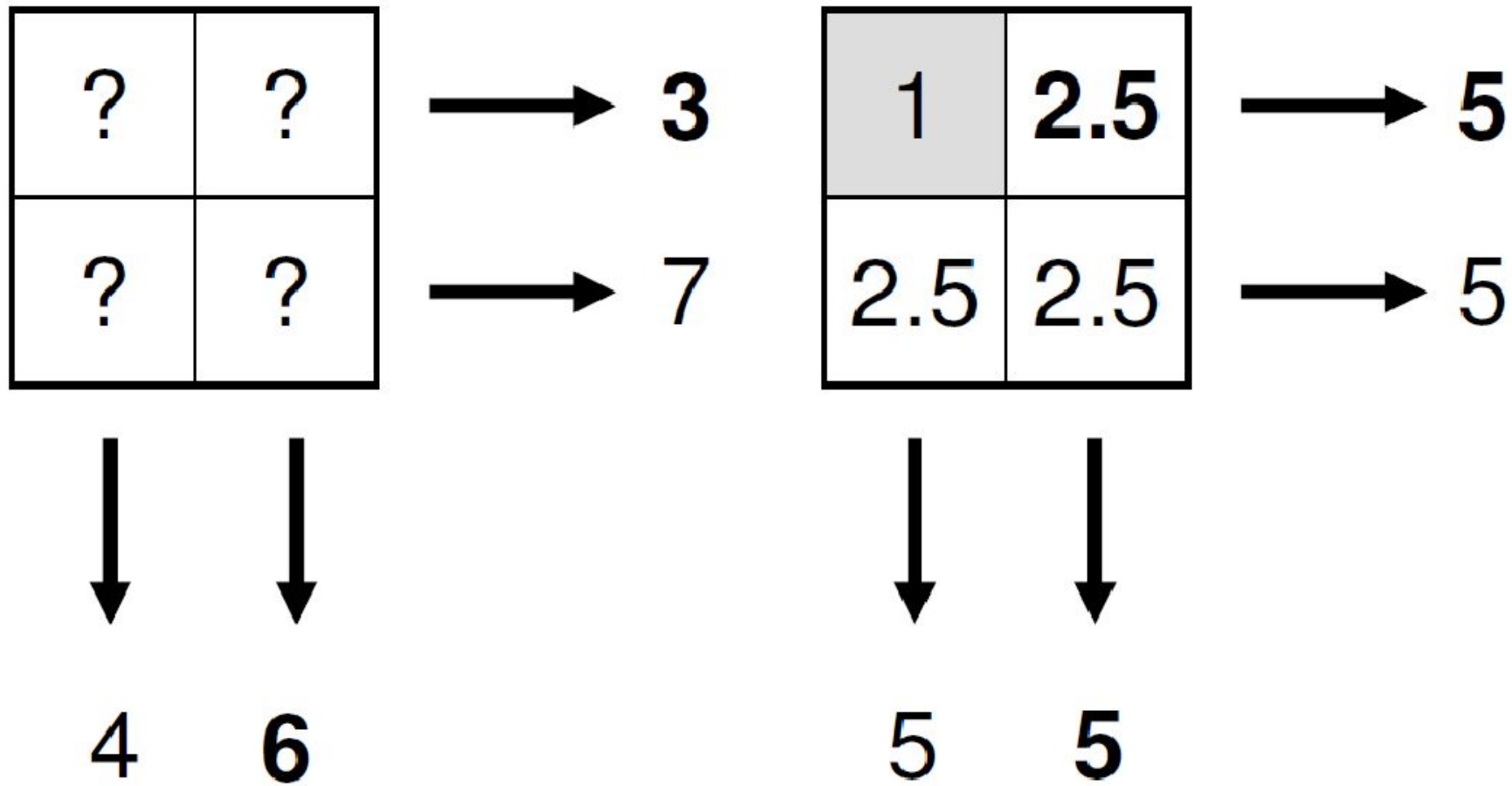
→ 5



5 5

$$c_{11} = (\mathbf{3} - \mathbf{5})/2 + (\mathbf{4} - \mathbf{5})/2 = -2/2 - 1/2$$

$$\mathbf{c_{11}} = -1 - 0.5 = \mathbf{-1.5}$$



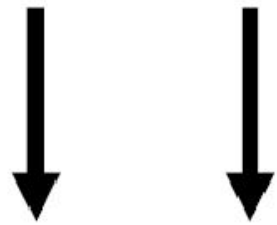
$$c_{12} = (3 - 5)/2 + (6 - 5)/2 = -2/2 + 1/2$$

$$c_{12} = -1 + 0.5 = -0.5$$

?	?
?	?

→ 3

→ 7

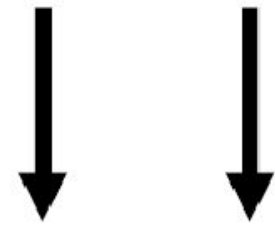


4 6

1	2
2.5	2.5

→ 5

→ 5



5 5

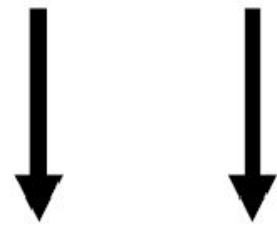
$$c13 = (7 - 5)/2 + (4 - 5)/2 = 2/2 - 1/2$$

$$c13 = 1 - 0.5 = 0.5$$

?	?
?	?

→ 3

→ 7

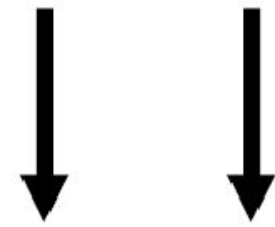


4 6

1	2
3	2.5

→ 5

→ 5



5 5

$$c14 = (7 - 5)/2 + (6 - 5)/2 = 2/2 + 1/2$$

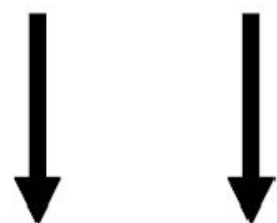
$$c14 = 1 + 0.5 = 1.5$$

Original data

1	2
3	4

→ 3

→ 7



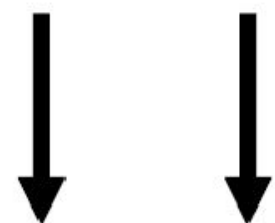
4 6

Reconstructed data

1	2
3	4

→ 5

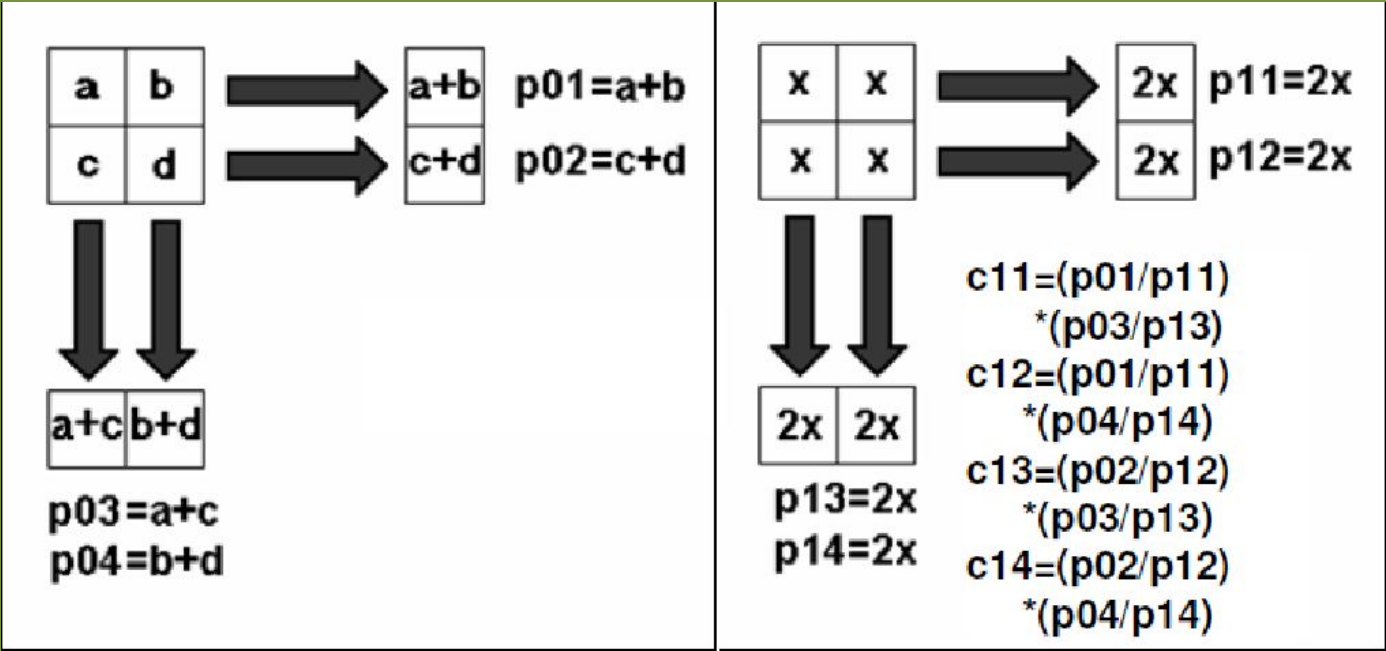
→ 5



5 5

A MULTIPLICATIVE RECONSTRUCTION TECHNIQUE

measured
projections

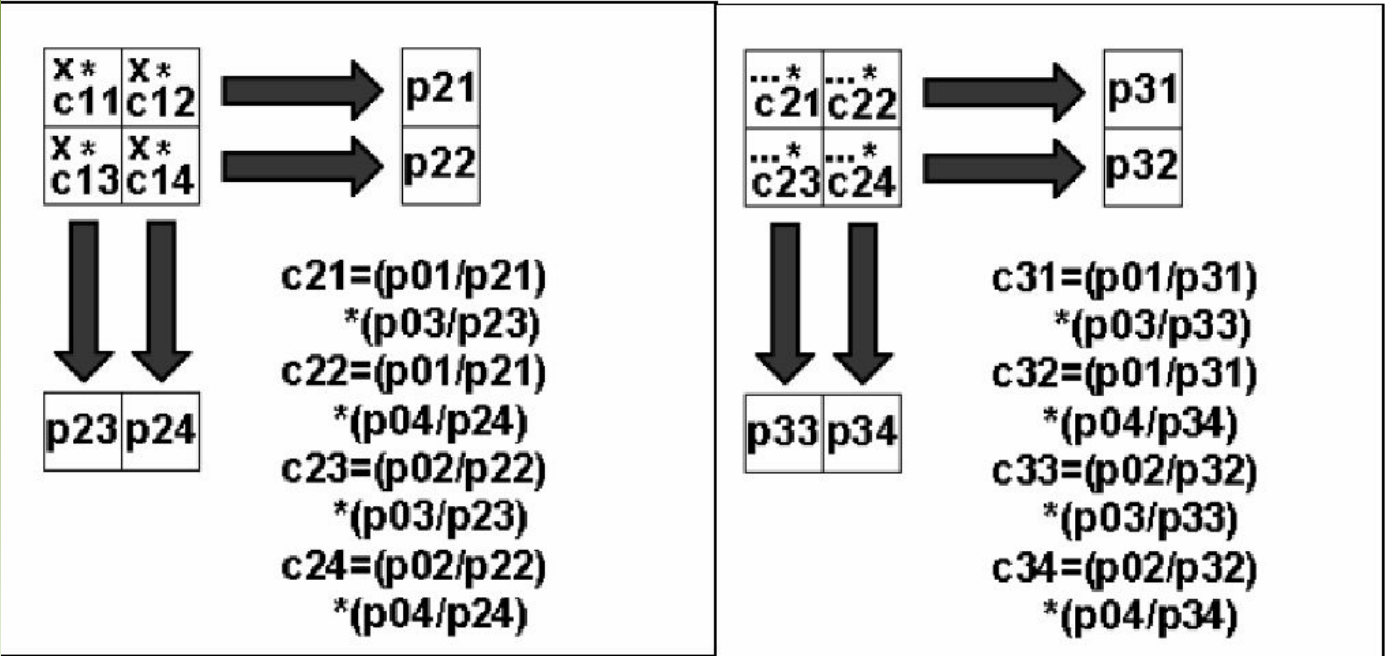


first estimate
and its
projections

$$x=(a+b+c+d)/4$$

correction
factors
(quotients
between
projections)

First
iteration
(multiplicative
corrections)



second
iteration
(multiplicative
corrections)